Fault signature comparison — KA15/KA22 (catastrophic failure) vs KA16/KA30 (success) vs K001 (healthy)

Time-domain (20 ms) + Hilbert envelope

FFT magnitude (0–8 kHz, log-y)

Envelope spectrum (0–1 kHz)

KA15 OR (real) (fail)

KA22 OR (real) (fail)

KA16 OR (real) (success)

KA30 OR (real) (success)

K001 Healthy (neutral)

time (ms)

frequency (Hz)

frequency (Hz)